

MRUNALI RAMESH NAVALKAR

Smart India Hackathon 2025 National Winner

• AI Engineer

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PROFESSIONAL SUMMARY

National Winner, Smart India Hackathon 2025 — selected out of 1,000+ competing teams across India.

Machine Learning & AI Engineering student with a proven track record of shipping intelligent systems — not just studying them. Built NLP pipelines at 85%+ accuracy, named entity recognition systems, and ML-based transformer diagnostics. Currently developing agentic AI workflows and RAG's with LangChain and LangGraph.

TECHNICAL SKILLS

Machine Learning: Scikit-learn · Feature Engineering · Exploratory data analysis · Data Preprocessing · Hyper Parameter Tuning

NLP : TF-IDF · Named Entity Recognition (NER) · Sentiment Analysis · Text Classification · Tokenization · NLTK

LLM Frameworks: LangChain · LangGraph · LangSmith · RAG

Python & Data: Python · Pandas · NumPy · Matplotlib · Seaborn · FastAPI

Tools & Databases: Docker · Git · SQL · Firebase · MongoDB · MLOPs

PROJECTS

AI-Driven Transformer Diagnostics System

**Smart India Hackathon
2025 — NATIONAL
WINNER**

Python · Machine Learning · Scikit-learn · Data Preprocessing · Matplotlib · CSV/XML

- Build an ML/DL-based fault detection system for power transformers using Frequency Response Analysis (FRA) data.
- Reduced manual diagnostic effort by automating multi-format data ingestion (CSV, XML) through a preprocessing pipeline that normalized and structured raw FRA datasets for ML model input.
- Improved fault detection reliability by applying machine learning pattern recognition to identify winding deformation and core faults with measurable accuracy gains over rule-based methods.
- Enabled faster maintenance decisions by developing interactive visualization dashboards that translated model outputs into actionable insights for field engineers.

NLP Pipeline — Text Classification & Sentiment Analysis

Python · NLP · NLTK · TF-IDF · Scikit-learn · Sentiment Analysis · NER · LangChain

- Achieved 85%+ classification accuracy on a real-world text dataset by building an end-to-end NLP pipeline using tokenization, stop-word removal, stemming, and TF-IDF vectorization.
- Improved information extraction quality by implementing Named Entity Recognition (NER) to identify and categorize entities from unstructured text across multiple document types.
- Reduced manual text processing steps by 60% by chaining NLP tasks into an automated pipeline using LangChain, enabling faster inference on new inputs.

- Validated model performance using precision, recall, and F1-score metrics across Naive Bayes and logistic regression classifiers to confirm generalization on unseen data.

Automated Smart Wheelchair System

Python • IoT • ESP32 • Firebase • React.js • Voice Control

- Improved mobility independence for users by integrating voice control, brain-signal input, and remote access into a single IoT-enabled wheelchair system.
- Achieved real-time responsiveness by connecting ESP32 sensors to a Firebase database and a React.js dashboard, enabling live monitoring with under 2-second latency.
- Designed the system around assistive technology principles, making the interface usable by individuals with limited physical mobility.

EDUCATION

Master of Computer Applications (MCA)

2024 – 2026 (Pursuing)

NCRD's Sterling Institute of Management Studies, University of Mumbai

Academic Topper

B.Sc. Information Technology

2024 • 8.88 CGPA

K.V. Pendharkar College, Dombivli, University of Mumbai

Available for AI/ML Roles & Internships • Open to Join Immediately